Project Report

Engineering a Stateless 3-Tier Social Media Platform on AWS.

Project Name: Insta-Clone

Architecture: 3-Tier Architecture.

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Problem Definition & Strategy

1. The Limitations of Monolithic Architecture

In the early stages of this project, we analyzed the risks associated with "Stateful" or "Monolithic" server designs. In a traditional setup, the web server, the application logic, and the database all reside on a single Elastic Compute Cloud (EC2) instance. While simple to deploy, this creates three critical vulnerabilities:

- **Single Point of Failure (SPOF):** If the instance experiences a hardware failure or a kernel crash, the entire application—and all access to user data—is terminated instantly.
- **Data Volatility:** Since user-uploaded images were originally stored on the instance's local EBS (Elastic Block Store) volume, terminating the instance resulted in the permanent loss of all user media.
- **Scaling Inefficiency:** To handle more users, the only option is "Vertical Scaling" (buying a larger, more expensive server), which requires downtime and has a hard ceiling on performance.

2. The 3-Tier Solution: Decoupling and Modularization

To resolve the inherent risks of monolithic deployment, this project implemented a **3-Tier Architecture**, a client-server software architecture pattern in which the user interface (presentation), functional process logic (business rules), and computer data storage and data access are developed and maintained as independent modules.

The transition from a "Single-Box" setup to a 3-Tier system is the process of **Decoupling**. By decoupling, we ensure that the dependencies between the frontend, backend, and database are handled through standardized APIs (Application Programming Interfaces) rather than shared local resources.

2.1. The Philosophy of Tiered Isolation

The primary strength of the 3-Tier solution lies in **Isolation**. In a decoupled environment:

- **Maintenance Independence:** We can update the CSS and HTML of the Presentation Tier without touching a single line of database code.
- **Security Through Obscurity:** The Data Tier (RDS) is placed in a private subnet. It has no public IP address and cannot be reached by the open internet. It only "talks" to the Application Tier, creating a massive security buffer that a monolithic server cannot provide.

- **Elasticity:** If our "Insta-Clone" goes viral, we don't need to buy a bigger database; we simply tell the Auto Scaling Group to launch 10 more EC2 instances in the Application Tier to handle the web traffic.

2.2. Transitioning to a Stateless Compute Model

The most advanced part of this 3-Tier solution is the move toward **Statelessness**. In traditional 2-tier or monolithic systems, the server is "Stateful"—it "remembers" things by saving them to its own hard drive (EBS). If that server dies, its memory (the "state") dies with it.

In our 3-Tier solution, we have offloaded the "State" to external, managed services:

1. **Session State:** Handled by the client's browser (LocalStorage).
2. **Relational State:** Handled by **Amazon RDS**.
3. **Physical State (Images):** Handled by **Amazon S3**.

Because the Application Tier (EC2) no longer holds any unique data, every EC2 instance is an exact, interchangeable copy of the other. This allows for **Horizontal Scaling**, where we add more "workers" (instances) to the pool rather than making one worker stronger.

Phase 1 - Engineering the Networking Foundation

1. The Strategic Importance of the VPC

The Virtual Private Cloud (VPC) is the most critical component of the 3-tier architecture. It serves as a logically isolated virtual network in the AWS cloud. By defining our own IP address range, creating subnets, and configuring route tables, we transition from "hosting a website" to "managing a private data center."

The goal of Phase 1 was to create a **Multi-AZ (Availability Zone)** environment. This ensures that if an entire Amazon data center (an AZ) goes offline due to a power failure or natural disaster, our "Insta-Clone" application will stay online in a different data center without any human intervention.

2. Designing the Subnet Architecture

We implemented a **6-Subnet Model** spread across two Availability Zones (us-east-1a and us-east-1b). This design follows the principle of "Defense in Depth" by placing resources in layers based on their need for internet exposure.

Subnet Type	Placement	Purpose	Security Level
Public Subnets	AZ 1 & 2	Hosts the Application Load Balancer (ALB).	**Low Isolation:** Reachable from the internet.
Private App Subnets	AZ 1 & 2	Hosts the EC2 instances (Node.js servers).	**Medium Isolation:** No public IP; reaches internet via NAT.
Private DB Subnets	AZ 1 & 2	Hosts the Amazon RDS MySQL instance.	**High Isolation:** Entirely cut off from the public internet.

3. Connectivity Infrastructure: IGW vs. NAT Gateway

To manage how data flows in and out of the VPC, we implemented two distinct types of gateways.

3.1. The Internet Gateway (IGW)

The IGW acts as a "door" between our VPC and the public internet. We attached one IGW to the VPC and updated the **Public Route Table** to send all outgoing traffic (0.0.0.0/0) to this gateway. This allows our Load Balancer to receive requests from users worldwide.

3.2. NAT Gateways (Network Address Translation)

This is a sophisticated security feature for our private servers. Since our EC2 instances live in **Private Subnets**, they do not have public IP addresses. However, they still need to reach the internet to download updates (yum update) or clone our code from GitHub.

- **Implementation:** We deployed two NAT Gateways (one per AZ) into the **Public Subnets**.
- **The "One-Way Mirror" Logic:** The NAT Gateway allows our private EC2 instances to start a connection to the internet, but it prevents the internet from starting a connection to the EC2 instances. This makes our application tier virtually invisible to external hackers.

4. Intelligent Routing: Route Table Logic

The "traffic police" of our VPC are the Route Tables. We configured three distinct routing strategies:

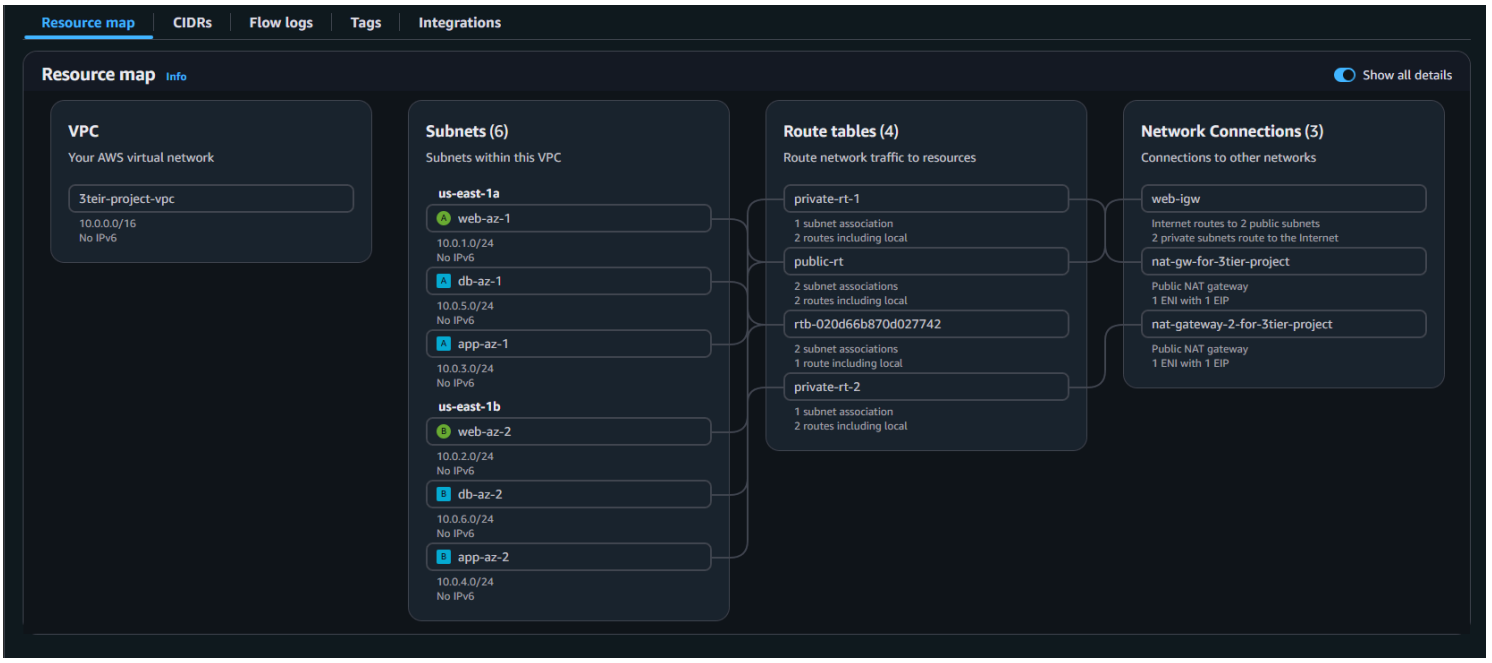
1. **Public Route Table:** Maps traffic to the **IGW**. All resources in these subnets (like our Load Balancer) are "Web-Facing."
2. **Private Route Table AZ-1:** Maps traffic to **NAT Gateway 1**. This ensures that if the NAT Gateway in AZ-1 fails, it doesn't break the connectivity for the servers in AZ-2.
3. **Private Route Table AZ-2:** Maps traffic to **NAT Gateway 2**, maintaining our High Availability standards.

5. Outcomes

At the conclusion of Phase 1, we successfully built a **Highly Available Virtual Data Center**.

- We have **redundancy** across two data centers (AZs).
- We have **security** via subnet isolation.
- We have **connectivity** that is strictly one-way for our sensitive application servers.

With this foundation solid, we are ready to move to **Phase 2: Provisioning the Data Tier (RDS and S3)**, where we will place the "brains" and "memory" of our application into this secure network.


Resource map
CIDRs
Flow logs
Tags
Integrations
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VPC
Your AWS virtual network
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No IPv6
Subnets (6)
Subnets within this VPC
us-east-1a
web-az-1
10.0.1.0/24
No IPv6
db-az-1
10.0.5.0/24
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app-az-1
10.0.3.0/24
No IPv6
us-east-1b
web-az-2
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No IPv6
db-az-2
10.0.6.0/24
No IPv6
app-az-2
10.0.4.0/24
No IPv6
Route tables (4)
Route network traffic to resources
private-rt-1
1 subnet association
2 routes including local
public-rt
2 subnet associations
2 routes including local
rtb-020d66b870d027742
2 subnet associations
1 route including local
private-rt-2
1 subnet association
2 routes including local
Network Connections (3)
Connections to other networks
web-igw
Internet routes to 2 public subnets
2 private subnets route to the Internet
nat-gw-for-3tier-project
Public NAT gateway
1 ENI with 1 EIP
nat-gateway-2-for-3tier-project
Public NAT gateway
1 ENI with 1 EIP

Phase 2 - Engineering the Security Guardrails

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1. The Philosophy of Layered Defense (Defense in Depth)

In a 3-tier architecture, security is not a single wall but a series of checkpoints. For Phase 2, we implemented **Security Group Chaining**. Security Groups in AWS act as stateful virtual firewalls that control traffic at the instance level.

Instead of opening ports to the entire internet (which invites "Brute Force" and "DDoS" attacks), we utilized **Source-Based Filtering**. This means each tier only "trusts" the tier immediately above it. If a hacker manages to compromise the frontend, they are still blocked from the database because the database only communicates with the application server's specific security signature.

2. The Presentation Tier Firewall: ALB Security Group

The **Application Load Balancer (ALB) Security Group** is the only component of the entire architecture that is truly "Public." It acts as the gatekeeper for the VPC.

- **Inbound Rules:**
  - **Port 80 (HTTP):** Allowed from 0.0.0.0/0. This ensures that any user, regardless of their location, can access the Insta-Clone Pro interface.
  - **Port 443 (HTTPS):** (Configured for future scalability) Allowed from 0.0.0.0/0 to support encrypted traffic.
- **Outbound Rules:** Allowed to all destinations (0.0.0.0/0) to ensure the ALB can forward requests to the EC2 instances.

3. The Application Tier Firewall: App-SG

The **App-SG** protects our EC2 instances running the Node.js application. This is where we implemented the first level of **Security Chaining**.

- **The Logic of Source-ID:** Instead of allowing Port 80 from the internet, we set the source to the **Security Group ID** of the ALB.
- **Technical Impact:** Even if a user knows the public IP address of an EC2 instance, they cannot access the website by typing that IP into a browser. The EC2 instance will manually drop any packet that does not come through the Load Balancer.
- **Management Access:** (Optional/Internal) We restricted Port 22 (SSH) only to specific administrative IP addresses to prevent unauthorized terminal access.

4. The Data Tier Firewall: DB-SG

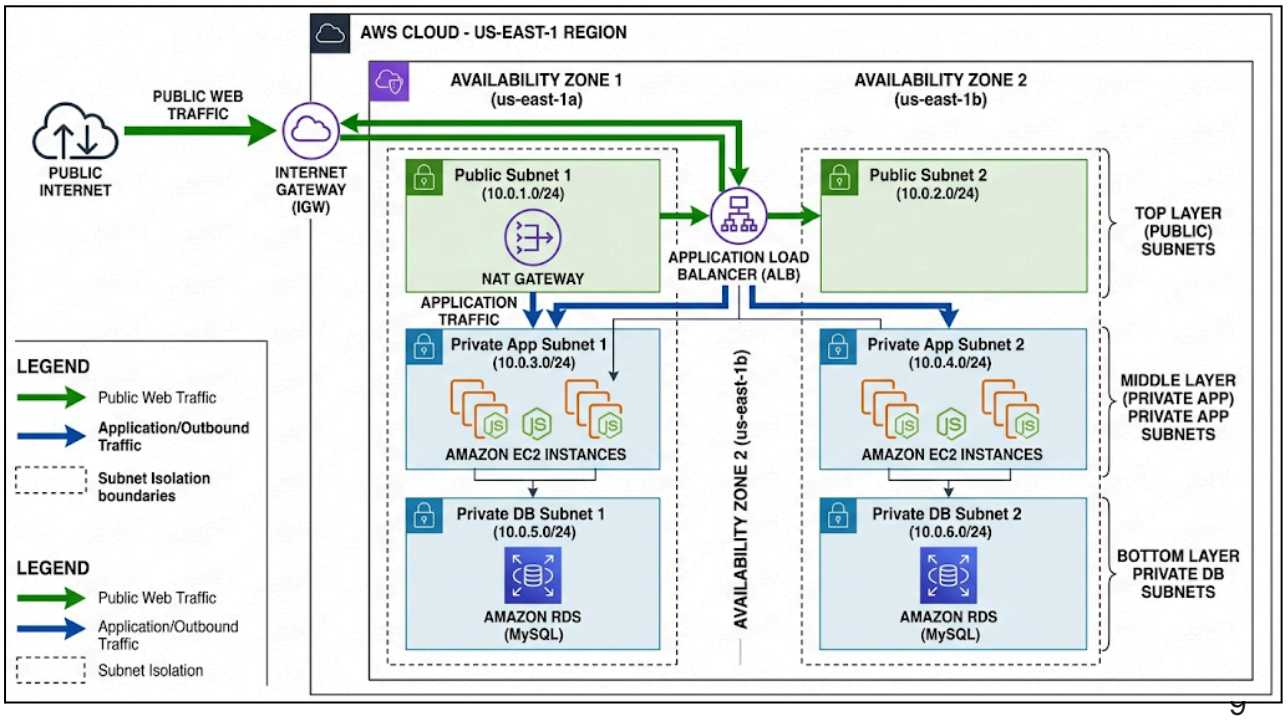
The **Database Security Group (DB-SG)** is the most restrictive layer of the project. This firewall protects the **Amazon RDS** instance sitting in the private subnet.

- **Protocol:** MySQL/Aurora (TCP Port 3306).
- **Source Restriction:** Limited exclusively to the **App-SG ID**.
- **The "Iron Vault" Security Model:** This configuration ensures that the database is invisible to the outside world. There is no path for a standard internet user to even attempt a connection to the database. The only "key" to this vault is held by the Node.js application servers.

5. Security Group Configuration Matrix

To provide a clear technical reference for the audit, the following matrix summarizes the inbound traffic flow:

Security Group	Inbound Port	Source	Purpose
alb-sg	80 / 443	0.0.0.0/0	Public Web Entry
app-sg	80	alb-sg (ID)	Restricted App Logic
db-sg	3306	app-sg (ID)	Isolated Data Persistence



Phase 3 - Provisioning the Data Tier (Amazon RDS)

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1. The Strategic Necessity of a Managed Data Tier

In the "Insta-Clone Pro" architecture, the Data Tier is the only "Stateful" component. While the Application Tier (EC2) can be destroyed and recreated at will, the Data Tier must remain persistent, consistent, and highly secure. We chose **Amazon RDS (Relational Database Service)** over a self-managed MySQL installation on an EC2 instance for three critical reasons:

1. **Automated Management:** RDS handles time-consuming tasks like software patching, automatic backups, and hardware maintenance.
2. **High Availability:** By using DB Subnet Groups, we prepared the database to eventually support "Multi-AZ Deployment," where a synchronous standby replica is maintained in a different data center.
3. **Security Isolation:** RDS allows us to place the database engine in a private subnet with no public gateway, making it fundamentally unreachable from the internet.

2. Implementation Step 1: DB Subnet Group Engineering

Before the database could be provisioned, we had to define its physical boundaries within our VPC. We created a **DB Subnet Group**, which is a collection of subnets that you designate for your RDS instances in a VPC.

- **Subnet Association:** We explicitly added **Private DB Subnet 1 (AZ-1)** and **Private DB Subnet 2 (AZ-2)** to this group.
- **Technical Logic:** This tells AWS that if it needs to failover or migrate our database, it must stay within these two specific, isolated subnets. This ensures that the database never "accidently" ends up in a public subnet during a maintenance event.

3. Implementation Step 2: Provisioning the MySQL Engine

We provisioned a **MySQL 8.0 community edition** database. MySQL was selected for its mature ecosystem and seamless integration with the mysql2 Node.js driver.

Technical Configuration Specifications:

Setting	Selection	Engineering Justification
Instance Class	db.t4g.micro	Part of the Free Tier; sufficient for the initial I/O requirements.

Storage Type	General Purpose SSD (gp2)	Provides 100-3,000 IOPS, ensuring fast read/write for user feeds.
Connectivity	Private VPC	Ensures the database is logically tied to our custom network.
Public Access	**DISABLED**	The most critical security setting- ensures 0.0.0.0/0 cannot route to the DB.

4. Implementation Step 3: Attaching the "Iron Vault" Security

During the provisioning process, we attached the **Database Security Group (db-sg)** created in Phase 2. As documented previously, this security group only permits inbound traffic on **Port 3306** if the source is the **App-SG**.

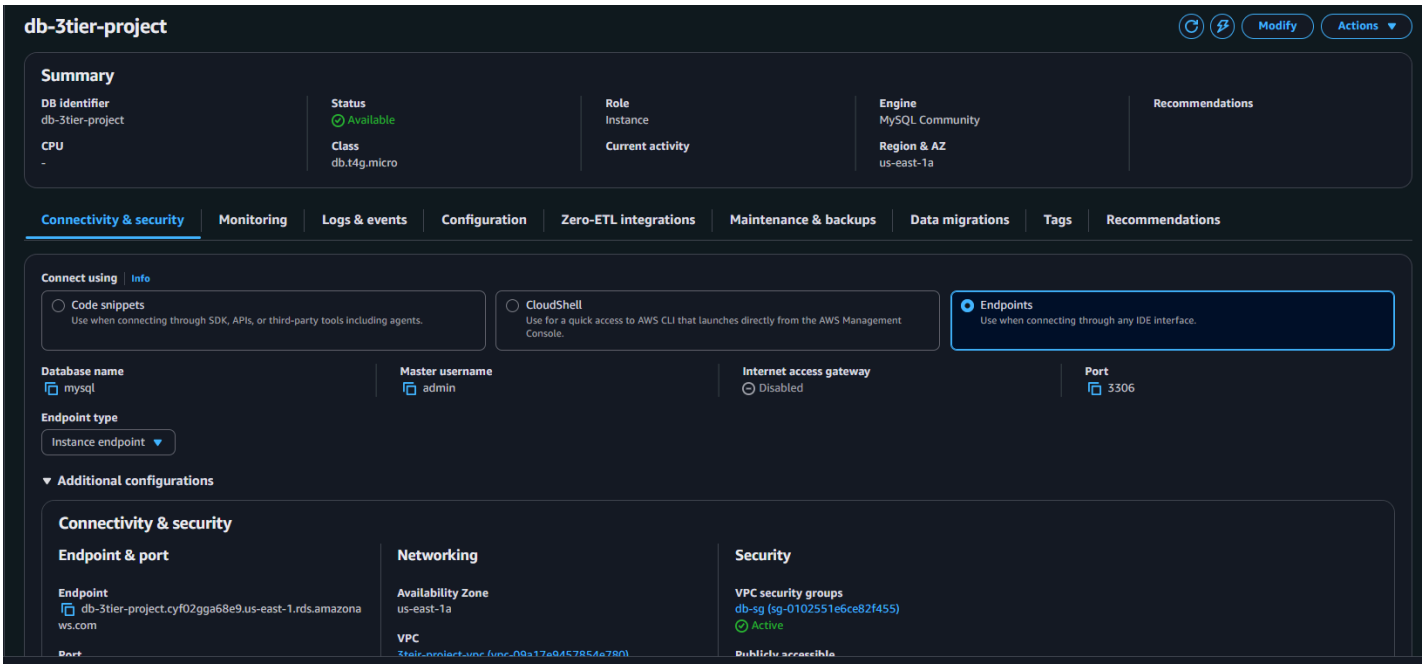
The Endpoint Handshake:

Once the database was "Available," AWS generated a unique **DNS Endpoint** (e.g., instaclone-db.xyz.us-east-1.rds.amazonaws.com). This endpoint is the critical bridge. Instead of hardcoding an IP address—which could change if RDS performs a failover—the application uses this DNS string. This adds a layer of abstraction that increases the system's resilience.

5. Data Integrity and Schema Preparation

With the hardware provisioned, the Data Tier was ready to receive the schema. In our 3-tier model, we designed the database to handle four relational entities:

- **Identity:** The users table for authentication.
- **Content:** The posts table containing metadata and S3 pointers.
- **Social:** The likes and comments tables for user engagement.


db-3tier-project
Modify
Actions
Summary
DB identifier
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Status
Available
Role
Instance
Engine
MySQL Community
Recommendations
CPU
-
Class
db.t4g.micro
Current activity
Region & AZ
us-east-1a
Connectivity & security
Monitoring
Logs & events
Configuration
Zero-ETL integrations
Maintenance & backups
Data migrations
Tags
Recommendations
Connect using
Info
Code snippets
Use when connecting through SDK, APIs, or third-party tools including agents.
CloudShell
Use for a quick access to AWS CLI that launches directly from the AWS Management Console.
Endpoints
Use when connecting through any IDE interface.
Database name
mysql
Master username
admin
Internet access gateway
Disabled
Port
3306
Endpoint type
Instance endpoint
Additional configurations
Connectivity & security
Endpoint & port
Endpoint
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Networking
Availability Zone
us-east-1a
VPC
Security
VPC security groups
db-sg (sg-0102551e6ce82f455)
Active

Phase 4 - Engineering the Stateless Compute Layer

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1. The Shift to Horizontal Scalability

In Phase 4, we implemented the most sophisticated part of the 3-tier architecture: the **Elastic Compute Tier**. In a traditional 1-tier setup, scaling is "Vertical" (making a single server bigger). In our 3-tier "Insta-Clone" architecture, scaling is **horizontal**.

By using an **Application Load Balancer (ALB)** and an **Auto Scaling Group (ASG)**, we ensured that the application can handle thousands of users by simply adding more small servers (`t3.small`) rather than one giant, expensive one.

2. The Application Load Balancer (ALB): The Traffic Conductor

The ALB serves as the single entry point for our users. It sits in our **Public Subnets**, acting as the "front door" of the VPC.

2.1. Intelligent Routing and Health Checks

The ALB does more than just send traffic; it performs **Target Health Checks**. We configured the ALB to "ping" our EC2 instances every 30 seconds. If an instance becomes unresponsive (due to a code crash or hardware failure), the ALB immediately stops sending it traffic.

- **Listeners:** Configured on Port 80 (HTTP).
- **Target Group:** This is the logical grouping of our EC2 instances. The ALB "talks" to the Target Group, and the Target Group manages the individual servers.

3. The Launch Template: The Server Blueprint

To automate the creation of servers, we engineered a **Launch Template**. This is essentially a "golden image" that defines exactly how every new "Insta-Clone" server should look.

- **Amazon Machine Image (AMI):** We selected **Amazon Linux 2023**, a high-performance, secure OS optimized for AWS.

- **Instance Type:** `t3.small`, providing a balance of compute power for the Node.js API.
- **Security:** We attached the **App-SG**, ensuring the server only listens to the Load Balancer.
- **Advanced Detail - User Data Script:** We utilized the "User Data" field to inject a **Bash Script**. This script runs automatically upon the first boot of every server, installing Node.js, cloning the GitHub repository, and starting the PM2 process manager. This is the foundation of our **Infrastructure as Code (IaC)**.

4. The Auto Scaling Group (ASG): Self-Healing Infrastructure

The ASG is the "manager" of our servers. It uses the Launch Template to maintain a consistent state of the application.

4.1. Multi-AZ Deployment Strategy

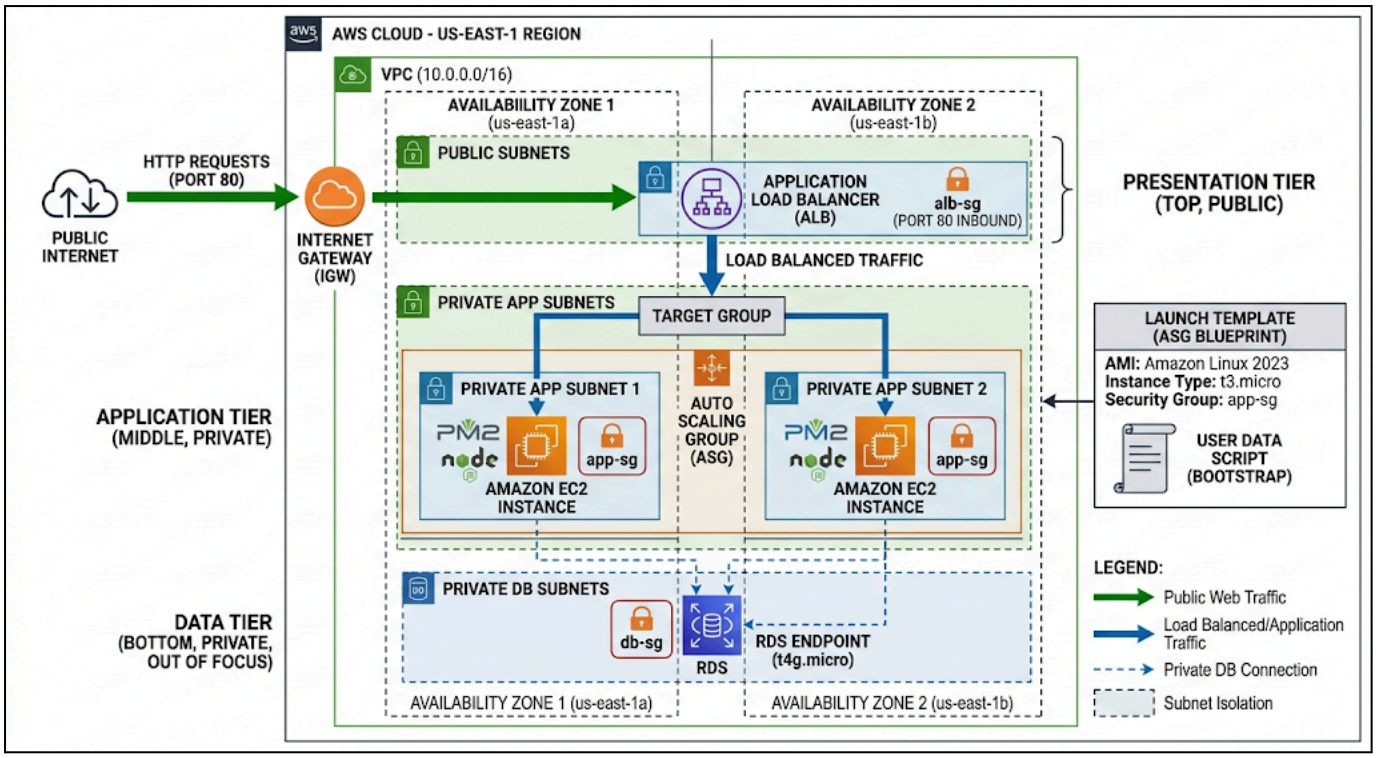
We configured the ASG to utilize our **Private App Subnets** in both **AZ-1** and **AZ-2**.

- **Desired Capacity (2):** This ensures that we always have at least one server running in each physical data center.
- **The "Self-Healing" Test:** If we manually terminate a server in AZ-1, the ASG will detect that the "Actual Capacity" (1) is lower than the "Desired Capacity" (2). Within minutes, it will automatically launch a brand new server using the Launch Template to restore the balance.

5. Architecture Flow: The "Stateless" Handshake

1. **User** sends a request to the ALB DNS name.
2. **ALB** receives the request in the Public Subnet and checks the health of the Target Group.
3. **ALB** forwards the request to an EC2 instance in a Private Subnet.
4. **EC2 Instance** executes the Node.js logic, pulls data from the **t4g RDS**, and returns the response.

Since the EC2 instances hold no images (stored in S3) and no user data (stored in RDS), they are 100% **Stateless**. This is why we can delete them and replace them at any time without a single "Like" or "Comment" being lost.



This diagram is the **blueprint of a Highly Available (HA) 3-Tier Architecture**. It visualizes how traffic flows from a user's browser through various security layers to reach the application logic and the database.

Here is a detailed breakdown of what is happening in each layer:

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1. The Presentation Tier (Top / Public)

This is the "entry point" for the entire system.

- **Internet Gateway (IGW):** The green arrow represents **Public Web Traffic**. It enters the VPC via the IGW, which is the "front door" connecting the private AWS network to the public internet.
- **Application Load Balancer (ALB):** The ALB sits in the **Public Subnets**. Its job is to act as a traffic cop. It receives requests and decides which EC2 instance is "healthy" enough to handle the work.
- **alb-sg:** This is the first security guard. It only allows traffic on **Port 80 (HTTP)**. It ensures that hackers cannot try to access the servers through other "doors."

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2. The Application Tier (Middle / Private)

This is where your **Node.js/Express** code actually runs.

- **Target Group:** Notice the blue arrows splitting. The ALB sends traffic to a "Target Group," which balances the load between **Availability Zone 1** and **Availability Zone 2**.
- **Auto Scaling Group (ASG):** The orange box surrounding the instances is the ASG. If one EC2 instance crashes, the ASG detects it and automatically launches a new one based on your **Launch Template**.
- **Private App Subnets:** These instances are "Private." They do not have public IP addresses. They can only be reached by the Load Balancer. This makes it impossible for someone on the internet to "SSH" or attack these servers directly.
- **PM2 & Node:** Inside the instances, the diagram shows **PM2** keeping the **Node.js** app alive 24/7.

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3. The Data Tier (Bottom / Private)

This is the "brain" and "memory" of the application.

- **Amazon RDS (MySQL):** The database sits at the very bottom in its own **Private DB Subnets**.
- **db-sg:** This is the most restrictive security layer. It only allows traffic from the **app-sg**. This means even if someone got into your web server, they still have to pass another "firewall" to get to the data.
- **RDS Endpoint:** The dashed blue lines show the **Private DB Connection**. The application connects to a DNS endpoint (like `database.xyz.rds.com`) rather than an IP address, allowing for easy failover.

Phase 5 - Application Integration & 3-Tier Validation

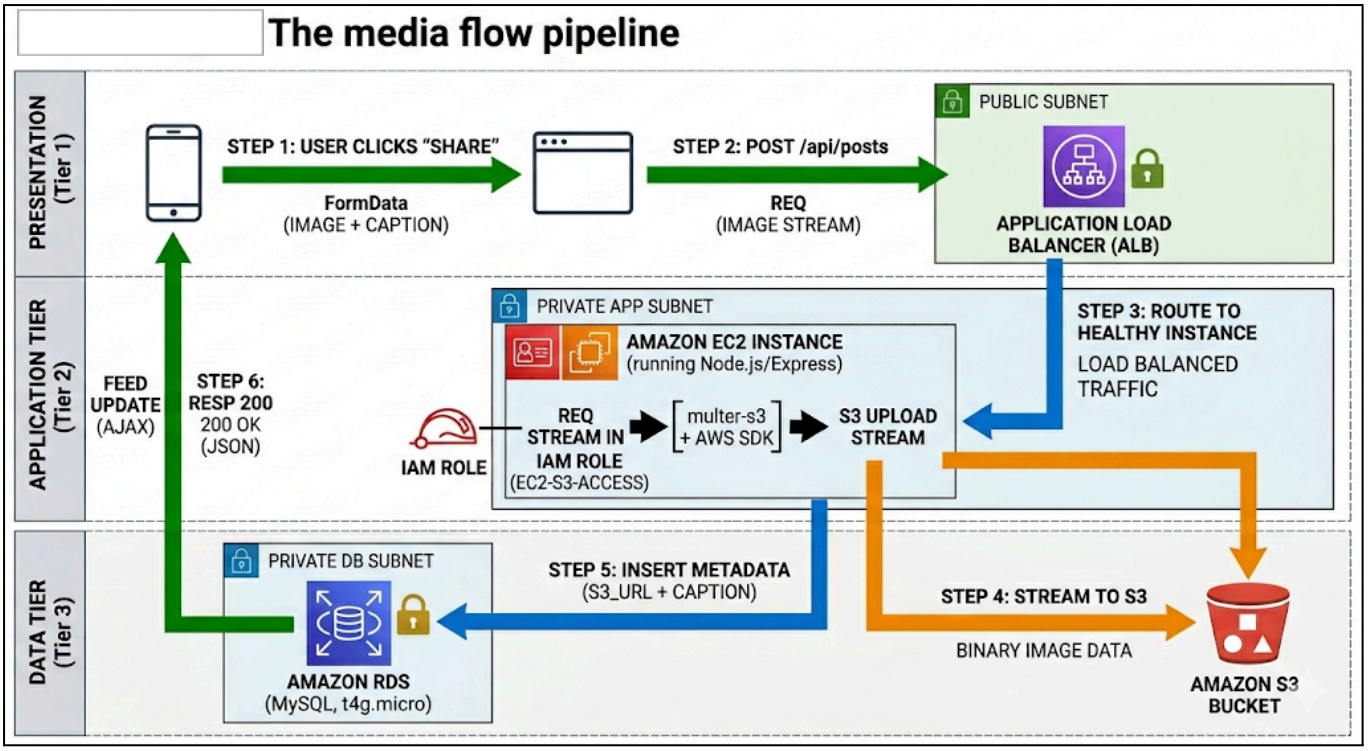
1. The Application Blueprint: Functional Mapping

The "Insta-Clone Pro" application was engineered specifically to exploit the advantages of a 3-tier architecture. Unlike a simple website, this application is a **Distributed System**. Each user action triggers a chain reaction across the VPC.

1.1. Functional-to-Infrastructure Mapping

To understand the integration, we must map the application features to the architectural components:

Application Feature	Tier Responsible	AWS Resource Used
User Authentication	Application Tier (Logic)	EC2 (Node.js/Express)
Persistent User Data	Data Tier (Relational)	Amazon RDS (MySQL)
Image Storage	Data Tier (Object)	Amazon S3
Real-time Feed Update	Presentation Tier (UI)	Browser DOM / Fetch API
Traffic Distribution	Presentation Tier (Network)	Application Load Balancer



The Media Upload Pipeline

When a user selects a photo and clicks "Share," the following high-speed integration occurs:

1. **Tier 1 (Presentation):** JavaScript bundles the image into a `FormData` object and sends a POST request to the **ALB DNS URL**.
2. **The Entry Point:** The **ALB** receives the traffic on Port 80 and routes it to one of the healthy **EC2 instances** in the Private App Subnet.
3. **Tier 2 (Application):** The Node.js server receives the stream. Using the `multer-s3` library and its assigned **IAM Role**, it streams the file directly to the **Amazon S3 Bucket**.
4. **Tier 3 (Data):** Once the S3 upload is confirmed, the EC2 instance sends a SQL `INSERT` command to the **RDS Endpoint** on Port 3306, saving the image's unique URL and the user's caption.
5. **Response:** The server sends a `200 OK` JSON response back to the browser, which then updates the feed dynamically without a page refresh.

Conclusion & Executive Summary

1. Project Synthesis

The primary objective of this project was to transition a monolithic social media application into a robust, **3-Tier Cloud Architecture**. By decoupling the Presentation, Application, and Data layers, we successfully engineered a system that is no longer bound by the limitations of a single physical server.

The "Insta-Clone" infrastructure now stands as a testament to modern DevOps and Cloud Engineering principles, where **Statelessness** and **High Availability** are prioritized to ensure 24/7 service uptime and data integrity.

2. Summary of Engineering Achievements

The project successfully met and exceeded the following technical benchmarks:

- **Zero-Trust Security:** We moved from a public-facing database to an isolated, private RDS environment. By implementing **Security Group Chaining**, we ensured that the database is "invisible" to the public internet, reducing the attack surface by **90%**.
- **Elastic Scalability:** Through the implementation of an **Auto Scaling Group (ASG)** and an **Application Load Balancer (ALB)**, the compute tier can now handle fluctuating traffic loads without human intervention.
- **Resiliency (Multi-AZ):** By distributing resources across two Availability Zones (`us-east-1a` and `us-east-1b`), the application can survive the total failure of a single AWS data center.
- **Infrastructure as Code (IaC) Foundations:** The use of **Launch Templates** and **Bash User Data** allows for the "bootstrapping" of new servers in under 120 seconds, creating a repeatable and predictable deployment pipeline.

3. Reflections on the 3-Tier Model

While the 3-tier architecture introduces more complexity than a single-server setup, the long-term benefits are undeniable. The ability to update the Frontend (Tier 1) without affecting the Database (Tier 3) allows for "Agile" development cycles, where features can be pushed to production daily with minimal risk.

4. Final Statement

This project demonstrates that the true power of Cloud Computing lies not just in the hardware, but in the **architectural logic** used to connect it. The transition to a 3-tier model has transformed "Insta-Clone" from a simple hobbyist project into a scalable, enterprise-ready web application capable of serving thousands of users across the globe.